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Filament conductivity enhancement through nonlinear beam interaction

DANIELLE REYES,[1,2] JESSICA PEÑA,[1] WIKTOR WALASIK,[3] NATALIA LITCHINITSER,[3] S. ROSTAMI FAIRCHILD,[1,4] AND MARTIN RICHARDSON[1,2,4,*]

[1] *Laser Plasma Laboratory, Townes Laser Institute, College of Optics and Photonics, University of Central Florida, Orlando, FL 32618, USA*
[2] *Physics Department, University of Central Florida, Orlando, FL 32618, USA*
[3] *Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708, USA*
[4] *Department of Physics and Space Sciences, Florida Institute of Technology, Melbourne, FL 32901, USA*
[*] *mcr@ucf.edu*

Abstract: Laser filament applications relying on filament plasma conductivity are limited by their low electron densities and corresponding short lifetimes. Filament plasma formation, an intensity-dependent process, is limited by the clamping of the filament core intensity. Consequently, increasing initial beam energy results in the breakup of the beam into multiple filaments rather than the enhancement of the electron density and conductivity of an individual filament. However, we demonstrate here the augmentation of the filament plasma density up to three times the typical value through the energy exchange between two co-propagating femtosecond beams with total powers between 1.7 and 2.2 P_{fil}.

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1. Introduction

Laser filamentation, i.e. the self-guiding of ultrashort laser pulses, continues to draw significant research interest both in understanding the fundamental physics of filament propagation and in realizing the many applications which laser filamentation affords. Such applications include kilometer-range LIDAR, remote THz generation, and free-space electromagnetic wave guiding [1,2]. A single laser filament forms as a result of the interplay between the Kerr self-focusing and plasma defocusing mechanisms present in the propagation of an ultrashort, high intensity beam in transparent media, such as atmospheric air. Above a certain critical peak power, P_{crit}=3.2 GW for λ_0=800 nm in air [2], the self-focusing overcomes diffraction, increasing the light intensity sufficiently to ionize the propagation medium through photoionization. The resulting plasma defocuses the trailing edge of the beam into a surrounding energy reservoir, which feeds energy into the beam core in a series of focusing-defocusing cycles [3], such that the beam maintains a high intensity core while producing a channel of plasma along the propagation axis. This process continues until the arrest of collapse. The interplay between these competing processes allows a filament to maintain a narrow beam diameter over an extended length and deliver high intensities over long distances. However, since self-focusing is balanced by plasma formation, it is worth noting that both the intensity carried by the filament and the plasma density are clamped to $\sim 10^{13}$ W/cm^2 and $\sim 10^{16}$ cm^{-3} [2], respectively.

The refractive index modification produced by a filament in air enables the use of filaments for guiding waves in the visible and IR, as well as microwaves [4–7]. To this end, multiple filaments can suitably be structured into an organized array, for instance, by manipulating their wavefront using phase plates [8–11]. The precise control of filaments and their arrangement in close proximity to effectively guide waves at long distances has prompted several studies into the interactions between multiple filaments [12–18]. Fluorescence measurements in studies of multi-filament interactions have indirectly shown changes in the plasma channel density and

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diameter due to the energy coupling that takes place between adjacent filaments [13]. A more recent study using interferometry found the peak density in a low numerical aperture (*NA*) filament bundle containing 5-8 filaments to be in the range of 9×10^{16} cm^{-3} in the focal region, though the interaction between neighboring filaments was not the focus of this study [19]. As a precursor to studying filament interaction in complex structures, much of the research has focused on the interaction between two co-propagating beams [20–33]. These studies have shown that two beams which are in phase may attract and fuse into a single filament or exhibit spiral motion, if they are propagating co-linearly or at a shallow crossing angle, respectively. Beams which are out of phase repel [20–25]. In addition to this, it has been shown that the outer energy reservoirs (each ~1 mm in diameter) of the beams [26] must overlap in order for energy coupling and interference to occur. Recent work found that two beams with powers below the filamentation threshold co-propagating with a separation of 330 µm formed a filament comparable to that created by a single beam with equivalent total energy [33]. No work has yet focused on the direct experimental characterization of the plasma produced by these interactions. Augmentation of the conductive properties of the channel due to filament interaction could substantially impact the structure and dynamics of filament arrays for guiding applications. In this work, we present direct measurement of the filament plasma dynamics produced by the interaction between two in-phase co-propagating beams with varying powers below and above the filamentation threshold.

2. Filament production and characterization methods

To perform this measurement, we used the Multi-Terawatt Filamentation Laser (MTFL) housed at the University of Central Florida, capable of delivering pulses with energies up to 500 mJ and pulse durations (τ) down to 45 fs at a repetition rate of 10 Hz and central wavelength of 800 nm. No more than 20 mJ were required per pulse for this experiment (τ=70 fs, 1/e^2 beam half-width w=8.5 mm). Interferometry was used to measure the electron density produced by filamentation [34–38]. This method does not require secondary calibration or additional information about the filamenting beam in order to determine plasma density and is sufficiently sensitive to detect the low densities of filament plasma. The interferometric measurement utilizes a pump-probe scheme in which 10% of the output beam energy was picked off to probe the filament and interfere with itself in a folded-wavefront interferometer (Fig. 1) [39,40], while the remaining 90% was split into two nearly identical beams and recombined in a beam combining interferometer [33]. This resulted in the loss of an additional 50% of the energy but enabled precise control over the spatial and temporal separation between the two beams used to produce filaments. A delay stage holding the mirrors in one arm of the beam combiner was adjusted to temporally synchronize the two beams while a translation stage holding the second beam splitter was adjusted to vary the initial beam separation and ensure parallel propagation.

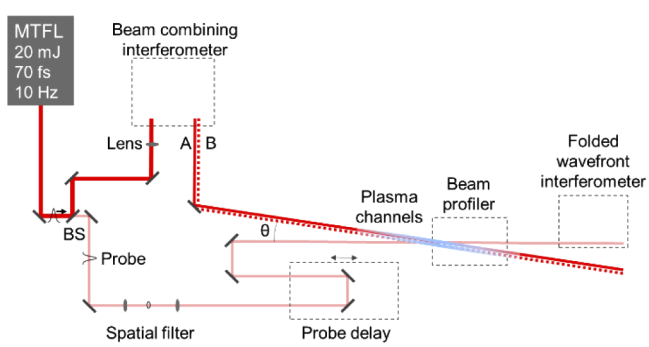

Fig. 1. Schematic of the system used for the measurement of the dual-beam filament plasma density and beam profile.

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Identical lenses in each arm of the beam combiner assisted the filament focusing. The lens focal length (f=2.5 m) was chosen to give an *NA* of ~0.0034, which corresponds to a regime where the nonlinear effects prevail in the filamentation process [41]. We recently reported on the notable difference between the optical structure formed when using tight focusing to induce filamentation (*NA*>0.0042), where linear effects dictate the focusing mechanics, as compared to loosely focused beams where nonlinear effects dominate and filamentation in the truest sense occurs [38,41]. Imaging of the filamenting beams was accomplished using a beam profiler, consisting of a series of wedges at grazing incidence and imaging optics [42]. From these images, the initial beam separation could be accurately tuned and the attraction and fusion of the beams visualized. To verify parallel propagation of the beams, the central location of beams A and B were measured individually and compared at various distances from the lens to ensure the separation between the independent propagation paths of the beams remained constant.

The probe beam was spatially cleaned and intersected with the filament at a shallow angle ($\theta < 1°$) in order to increase the interaction length, and thus sensitivity, to the small phase perturbations produced by the weak filament plasma. The probe then entered the folded wavefront interferometer, producing interferograms, from which the phase shift could be extracted using Fourier analysis. Abel inversion of the phase lineout yielded the refractive index change (Δn) produced by the filamenting beam, from which electron density (ρ) could be easily calculated (Δn=-$\rho/2\rho_c$) [38]. To capture the plasma dynamics, adjustable rail mounted mirrors were used to change the delay time of the probe relative to the filamenting beams. Plasma dynamics were studied for three different energy distributions: (1) two non-filamenting beams, (2) one non-filamenting beam with one filamenting beam, and (3) two filamenting beams. For case 2, the imperfect splitting ratio of the 50:50 beam splitters (~55:45) was used to produce an overall factor of ~1.4 difference in energy between the two output beams. These same beam splitters were used for case 1, but could not be used for the final high energy case due to the thickness of the glass. For case 3, these were replaced by ultra-thin beam splitters in order to reduce the amount of accumulated B-integral and prevent the early onset of filamentation which would be damaging to the optics. For each case, the initial separation between the beams was chosen to be a value <500 μm to ensure beam interaction, which requires the overlap of neighboring energy reservoirs. Therefore, separations of 180 ± 40 μm and 280 ± 40 μm were used. In all the cases, beams A and B were measured independently in addition to the dual-beam filament using the system shown in Fig. 1.

The filamentation threshold energy under these experimental conditions was determined to be ~3 mJ or a power of P_{fil}~40 GW. The value of P_{fil} is known to be a factor of $\pi^2/4$ (~2.5) times P_{crit} [43]. In addition, the power required for a pulse to filament may exceed this value of P_{fil}, depending on the coupling of pulse energy into the filament, which varies with beam diameter and convergence (optimal for w<1.5 mm and NA<0.002) [14]. It is important to recognize that the energy coupled into the filament remains independently limited to ~1mJ despite slight variations in input beam power [2,44]. The non-filamenting beams were given energies slightly below the filamentation threshold, with corresponding powers lying between $0.5P_{fil}$ and $0.9P_{fil}$. The filamenting beams were given energies above the filamentation threshold, corresponding to powers between $1.0P_{fil}$ and $1.4P_{fil}$. Across all the data, the average value of the peak plasma density for a single-beam filament containing these powers was measured to be 1.3×10^{16} cm^{-3}, with a standard deviation of $\sigma = 0.2\times10^{16}$ cm^{-3}. It should be noted that the results showed no trend of increasing density with increasing power as expected from the phenomena of intensity clamping. One recent study similarly demonstrated that the filament plasma density showed no increase within error despite a near doubling of the input power from $1.0P_{fil}$ to $1.7P_{fil}$ [38]. Therefore, this average value will be used throughout the paper for comparison with standard filamentation.

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3. Beam interaction studies

3.1. Dual-beam plasma density and decay time enhancement

Due to the high intensities in a filament, using two subthreshold beams to form a single filament has advantages from the standpoint of filament beam engineering [33]. Two subthreshold beams were previously seen to fuse into a single filament for separations less than 330 µm [33]. Therefore, beams A ($0.99P_{fil}$) and B ($0.69P_{fil}$) were initially given subthreshold powers so that independent filamentation did not occur, and the interaction of the two non-filamenting beams was studied for a separation of 180 µm. Interferometric measurement of the individual beams confirmed that filament-level plasma densities were not produced, since any plasma production fell below the detection limits of our system (minimum detection level of $\sim 1 \times 10^{15}$ cm^{-3}). The filament beam profiler was used along propagation to confirm the parallelism of the beam paths when not interacting and the merging of the beams into a single hotspot when co-propagating in-phase with one another. After the merging of the two subthreshold beams containing a total power of $1.7P_{fil}$, plasma formation was detected (Fig. 2, blue circles). Each data point plotted in Fig. 2 represents the average and standard deviation of the electron density over 100 laser shots. The peak plasma produced by the merging of the beams was 5σ greater than that formed by a single beam or an ~80% increase in peak density.

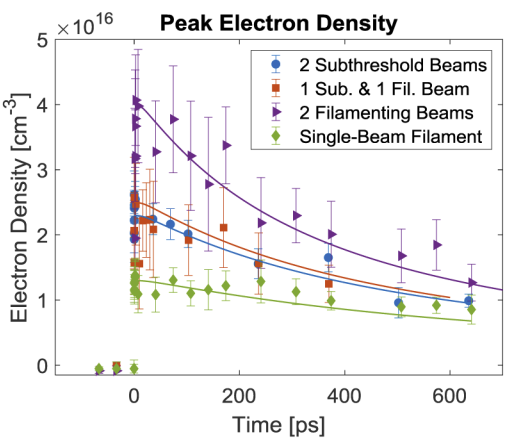

Fig. 2. The on-axis electron density values for the dual-beam filaments for each energy combination considered with a separation of 180 µm between beams A and B. An example single-beam filament is plotted with green diamonds for reference.

The decay of the plasma was simulated using an air plasma chemistry model proposed in [38], plotted as the solid lines in Fig. 2. The initial electron density for the simulation was taken from the experimental data. From this simulation, the temporal duration of the plasma was determined in order to compare the relative persistence of the plasma between the single and dual beam filament cases. Because the plasma channel persists beyond the half-life (a quantity which is known to decrease with increasing peak density, as a result of initial rapid decay [3]), characterization of the relative survival time of the plasma was determined using a threshold density value of 5×10^{15} cm^{-3} . For the remainder of this paper, the plasma survival time will be defined as the time required for the density to drop below this threshold. This choice in threshold value is such that the survival time is equivalent to the half-life for a filament with the typical density of 1×10^{16} cm^{-3}. For all the data, the average survival time of a single-beam filament was 0.98 ± 0.13 ns (Table 1). By comparison, the dual-beam plasma resulting from the merging of two subthreshold beams (Fig. 2, blue circles) was observed to survive for ~0.4 ns longer.

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Table 1. Peak electron densities, measured and modeled, and survival times for the dual-beam plasmas for all energy distribution conditions and a beam separation of 180 µm. The top row gives these values for a single-beam plasma for comparison.

Beam(s)	Density ($\times 10^{16}$ cm^{-3})	Survival (ns)	Model Density ($\times 10^{16}$ cm^{-3})
1 filamenting beam	1.3 ± 0.20	0.98	–
2 subthreshold beams	2.3 ± 0.20	1.337	**2.14**
1 sub. & 1 fil. beam	2.5 ± 0.54	1.378	3.32
2 filamenting beams	4.0 ± 0.50	1.565	*3.88*

In the second energy distribution considered, beams A and B carried powers above ($1.26P_{fil}$) and below ($0.91P_{fil}$) threshold respectively (Fig. 2(b), orange squares). The electron densities produced by beam A independently and by both beams combined were measured. The introduction of beam B brought the total power to $2.2P_{fil}$. Typically, increasing input power in a single beam beyond $2P_{fil}$ results in the breakup of the beam into approximately $N \sim P_{in}/P_{fil}$ filaments [2,43]. However, this was not the case for the interacting beams. Instead, the introduction of beam B doubled the peak plasma density of the filament compared to that produced by beam A alone. In addition, it was previously shown that an increase in single pulse power by a factor of 1.7 did not enhance plasma density [38]. However, an identical increase in total power here resulted in density enhancement due to the beam interaction. The resulting dual-beam plasma was comparable to that formed by the fusion of two subthreshold beams with total power of $1.7P_{fil}$. The plasma again survived for ~0.4 ns longer than that generated by a single beam.

For the final case considered, beams A ($1.1P_{fil}$) and B ($1.0P_{fil}$) both carried powers sufficient to form filaments. Despite the total dual-beam power exceeding $2P_{fil}$, the interacting beams again did not break up into two filaments as would be expected for a single beam with equivalent power. Instead, the beams fused into a single filament with higher plasma density than a single-beam-induced plasma. With a total power of $2.1P_{fil}$ (Fig. 2, purple triangles), the density was increased by ~200% compared to the single-beam filament densities (1.3×10^{16} cm^{-3}). The plasma survival time likewise exceeded that of a typical filament by ~0.6 ns.

Data was also taken for the same three energy distribution cases but with an initial separation of 280 µm (not shown in this paper). These results followed a similar trend of increasing electron density with the transition from cases 1 through 3. However as would be expected, the enhancement was less significant than that observed at a separation of 180 µm.

3.2. Theoretical plasma properties in nonlinear beam interaction

The experimental results in this work were reproduced by numerical simulations. The evolution of the slowly varying envelope of the electric field of a high-power, ultrashort pulse in air can be described by the nonlinear Schrödinger equation [2,17] and the parameters used here are the same as in Ref. [33]. The numerical simulations of the propagation of the filaments were performed using a (3 + 1) dimensional split-step Fourier method. The results of the simulations are shown in Table 1. The energies of the beams used in the numerical simulations were chosen in such a way so that the peak electron densities generated by the filaments for single beam experiments were accurately reproduced. The discrepancy in the energies arises from a difference between the theoretical and experimental critical power for self-focusing [41]. While typically assumed to be around 3.2 GW [2], the critical power has been shown to increase for pulse durations less than 200 fs, reaching as high as 10 GW for a 42 fs pulse [45], necessitating a lower input pulse power for the simulation compared to experiment. Once the appropriate energies were selected for the filamenting beams, the initial separation distance between the two beams was set to 150 µm, at which separation the electron density created by the combination of the beams matched the values obtained in the experiment (italic entry in Table 1). This separation distance was

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slightly smaller (by ~17%) than that used in the experiment but was within the experimental error. This separation distance was maintained for the remainder of the numerical simulations. The energy of the beams was then reduced to just below the filamentation threshold, to produce negligible electron densities and at the same time replicate the electron densities obtained for the case of the subthreshold dual-beam experiments (see bold entry in Table 1). Finally these same parameters were used in the dual-beam experiment with one subthreshold and one filamenting beam (underlined entry in Table 1). In this case, the electron density obtained in the numerical simulation was within 1.6σ of the experimental result ($\sigma = 0.54\times10^{16}$ cm^{-3}).

Two simulation methods for the plasma density evolution were compared: (i) numerical solution of rate equations taking into account the air chemistry [38] and (ii) analytical formalism provided in Ref. [2]. The system of coupled equations governing the evolution of electron density and densities of ions and neutral particles has an analytical solution [2] that, in the absence of electric field (after the laser pulse has passed and there is no external electric fields), can be approximated as:

$$\rho(t) = \rho_0/(1 + \rho_0\beta_{ep}t), \tag{1}$$

where $\beta_{ep} = 1.2\times10^{-7}$ cm^3/s denotes the electron-ion recombination rate [46] and ρ_0 is the initial electron density. The shape of the decay curve using the approximate form of the analytic solution did not follow the data quite as well as the air chemistry simulation at high initial densities but was similar for low densities (Fig. 3(a)). The decay predicted by the equation is initially more rapid and then becomes slower than the air chemistry simulation. A comparison of the predicted survival times for the measured peak densities is shown in Fig. 3(b).

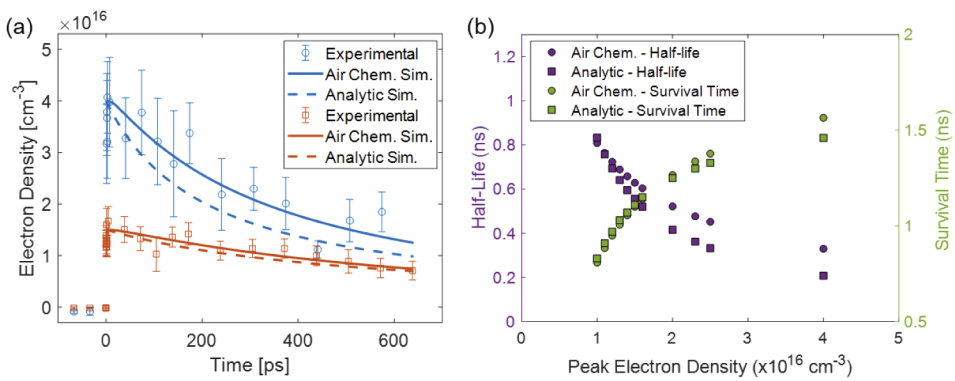

Fig. 3. (a) Comparison of air chemistry (solid line) and analytic (dashed line) simulations to experimental data for high (circles, blue) and low (squares, orange) peak densities. (b) Comparison of relative half-life (left, purple) and survival time (right, green) derived from air chemistry (circle) and analytic (square) simulations for each measured peak density.

3.3. Additional beam characterization

To further explore the cause of the plasma enhancement, the beam was characterized by other diagnostics in conjunction with electron density measurements. For each energy condition and beam separation, beam profiles of the individual beams and the corresponding combined beam were obtained as a means of fluence profile measurement (Fig. 4). The data shows that the filaments resulting from the recombined beams contained peak fluences that were from 1.8 to 2.6 times that of the filaments produced by individual beams (~2.1 J/cm^2). The beam profiles also showed that the full-width-at-half-maximum (FWHM) diameters of the average dual-beam and average single-beam filaments were ~210 ± 40 μm and ~200 ± 30 μm respectively, with no pattern of change in filament core diameter caused by the merging of the beams. Interferometric

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measurement of the refractive index change was also performed at times prior to plasma formation, at which point $\Delta n = n_2 I$. Using n_2=5.57×10^{19} cm^2/W [33], the average peak filament intensity was determine to be ~1×10^{13} W/cm^2. Of greater importance to this study than the peak value of the intensity was the relative change in filament intensity. Little to no increase in filament intensity was seen to result from the beam combination. The average dual-beam filament intensity was only a factor of 1.1 times the average single-beam filament intensity, which was within the error. These findings imply an increase in the energy coupling into the filament core and a simultaneous extension of the pulse duration resulting from the beam combination, thus maintaining similar peak intensities and core diameters while increasing the fluence. This more energetic longer duration pulse would then ultimately be responsible for the enhanced electron density.

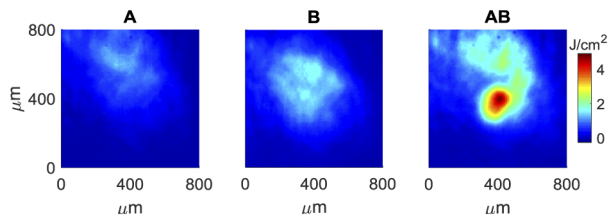

Fig. 4. Beam (fluence) profiles of beam A, beam B, and beams A&B for the case of two subthreshold beams forming a filament, separated by 180 μm.

4. Discussion

A balance between the effects governing filamentation, primarily the Kerr effect and plasma defocusing, results in a stable, clamped intensity in the filament core. Therefore, the fact that little increase in intensity was seen here is expected. Since the filament core diameter was likewise unaffected by the dual beam interaction as recorded in the beam profiles, an increase in the amount of energy in the filament core and a simultaneous increase in pulse duration would account for the slight increase in peak intensity (~10%) while simultaneously doubling or tripling the peak beam fluence. It is known that the nonlinear processes involved in filamentation change the pulse duration [2]. Both the slight increase in intensity and the extended pulse duration contribute to the resulting increase in electron density. Previous simulations of two filaments merging at a shallow angle made by Kosareva, *et al.*, [47] predicted a doubling of the electron density with a corresponding increase in peak intensity by only 30%, and the enlarged density was attributed to an increase in pulse duration due to the merging of the filaments. Here, we confirm that the filament plasma resulting from the merging of two beams has a greater electron density than that from single beam filamentation, for an initial beam separation of ~180 μm. For a filament resulting from the merging of two filamenting beams containing $2.1P_{fil}$, the plasma density is three times the typical filament density.

It should be noted that, under normal conditions, it is not possible to significantly increase filament plasma density by arbitrarily increasing the energy in the filament in the nonlinear propagation regime [38]. A beam with initial power many times the critical power will break up into several filaments, each with an intensity of ~10^{13} W/cm^2, due to modulational instability [1]. In a single beam possessing sufficient energy, irregularities in the wavefront lead to further distortions and hot spots due to nonlinear effects [11], causing the beam to break up. While increasing the energy and inducing multi-filamentation does increase the overall production of electrons, the volume occupied by those electrons is likewise increased so that there is no enhancement in the electron density. Along propagation, these individual filaments may interact and merge leading to the type of enhancement seen here. However, there is no control over this process, with the multi-filamentation pattern changing from shot-to-shot, and no guarantee

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this merging and enhancement will occur. By allowing two beams to interfere prior to the accumulation of large B-integral, the wavefront is engineered to have one dominant hot spot. This engineered hot spot will produce a filament with threefold increase in electron density and 1.6 times longer survival time compared to a single-beam filament, likely due to a longer pulse duration in addition to higher core and reservoir energy. With this method, the electron production is increased while maintaining the same volume as a single filament. Comparing a single filament to a single wire transmitting microwaves at a frequency of 10 GHz, this augmentation of the plasma density corresponds to an enhancement in conductivity from 3.7 to 11.3 $\Omega^{-1}\text{cm}^{-1}$, using $\sigma_{pl}=\omega_{pl}{}^2\nu_c\varepsilon_0/(\omega_{mw}{}^2+\nu_c{}^2)$, where $\omega_{pl}=(\rho e^2/m_e\varepsilon_0)^{1/2}$ is the plasma frequency, $\nu_c=1\times10^{12}$ Hz is the effective rate of elastic collisions of electrons and neutrals, ε_0 is the permittivity of free space, e and m_e are the charge and mass of an electron, and ω_{mw} is the transmitted microwave frequency [5]. Based on these results, arrays of filaments could be engineered to position neighboring pairs of filaments within <500 μm of each other in order to take advantage of the attraction, merging, and resulting enhancement in plasma density and lifetime. Similarly, a phase mask producing structured arrays of subthreshold beams could be co-propagated with a second identical array in order to produce filament arrangements while curtailing risk of damage to optics.

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Disclosures

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